North Lake Senior Campus

ANSWERS

CHEMISTRY 12 SEMESTER 1 EXAMINATION 2014

Multiple Choice: Mark the answer by shading in the box to the LEFT of the letter you chose.

1. ☐a ■b ☐c ☐d

2. ■a ☐b ☐c ☐d

3. ☐a ☐b ■c ☐d

4. ■a ☐b ☐c ☐d

5. ☐a ■b ☐c ☐d

6. ☐a ■b ☐c ☐d

7. ☐a ■b ☐c ☐d

8. ■a ☐b ☐c ☐d

9. ☐a ☐b ☐c ■d

10. ☐a ☐b ☐c ■d

11. ☐a ☐b ■c ☐d

12. ■a ☐b ☐c ☐d

13. ☐a ■b ☐c ☐d

14. ■a ☐b ☐c ☐d

15. ☐a ☐b ☐c ■d

16. ☐a ■b ☐c ☐d

17. ☐a ☐b ☐c ■d

18. ☐a ☐b ■c ☐d

19. ☐a ☐b ☐c ■d

20. ☐a ☐b ☐c ■d

21. ☐a ☐b ■c ☐d

22. ☐a ☐b ☐c ■d

23. ☐a ■b ☐c ☐d

24. ☐a ☐b ☐c ■d

25. ☐a ■b ☐c ☐d

Section Two: Short answer 41% (74 Marks)

This section has **11** questions. Answer **all** questions. Write your answers in the space provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time for this section is 70 minutes.

Question 26

Write ionic equations for any reactions that occur in the following procedures.
If no reaction occurs, write "no reaction".
In each case describe in full what you would observe, including any colours, odours, precipitates (give colour), gases evolved (give colour or describe as colourless).
If no change is observed, you should state this.

(a) Iron (III) chloride solution is added to potassium hydroxide solution.

Equation: $\mathbf{Fe^{3+}(aq) + 3\ OH^{-}(aq) \rightarrow Fe(OH)_3(s)}$ (2)

Observation : **From a clear and brown solution a brown solid forms.**

(1)

(b) Dilute hydrochloric acid is poured onto solid magnesium hydrogen carbonate:

Equation: $\mathbf{2H^{+}(aq) + Mg(HCO_3)_2\,(s) \rightarrow Mg^{2+} + 2CO_2(g) + 2H_2O(l)}$

(2)

Observation: **White solid dissolves, colourless gas evolved from colourless solution.** (1)

(c) A piece of magnesium ribbon is added to a solution of ethanoic acid.

Equation: $\mathbf{2CH_3COOH_{(aq)} + Mg_{(s)} \rightarrow 2CH_3COO^{-}_{(aq)} + Mg^{2+}_{(aq)} + H_{2(g)}}$

(2)

Observation: **Grey solid dissolves to form colourless solution. Colourless, odourless gas produced. (Vinegar smell dissipates)**

(1)

Question 27

Write the symbol for (i) an element, and (ii) an ion which have the electron configuration shown

	electron configuration	element	A monatomic ion with
(a)	2 8 8	**Argon**	a positive valency: **K^+ or Ca^{2+} or Sc^{3+}**
(b)	2 8	**Neon**	a negative valency: **F^- or O^{2-} or N^{3-}**

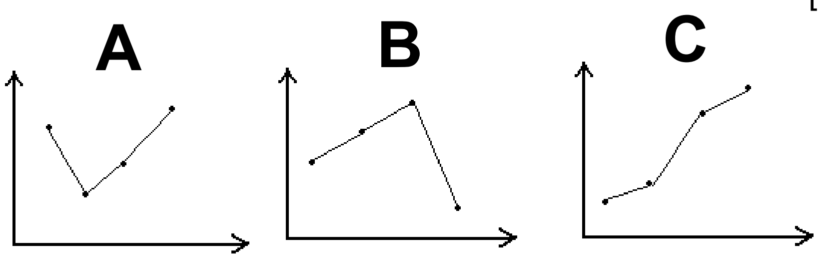
[4 marks]

Question 28

For each of the unlabelled graphs below select which one best represents the expected trend for:

- Boiling Points of the hydrides of group 16 **A**.
- Melting Points of elements Atomic numbers 12, 13, 14, 15. **B**.
- First Ionisation Energy values of elements 12, 13, 14, 15. **C**.
- Successive Ionisation Energy values for a group 2 element. **C**.

[4 marks]

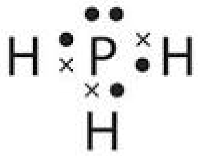


Print the letter corresponding to the graph on the line alongside the description above.

Question 29

For each species listed in the table below, draw the electron dot diagram, representing all valence shell electron pairs either as **:** or as — **and** state or draw the shape of the molecule.
In the last column, indicate whether the substance is polar or non-polar.

for example, water $\mathrm{H}:\ddot{\underset{\cdot\cdot}{\mathrm{O}}}:\mathrm{H}$ or $\mathrm{H}{-}\ddot{\underset{\cdot\cdot}{\mathrm{O}}}{-}\mathrm{H}$ or $\mathrm{H}{-}\overline{\underline{\mathrm{O}}}{-}\mathrm{H}$ Bent Polar

Molecule	Structural formula (showing all valence shell electrons)	Shape (sketch or name)	Polar or Non-polar
phosphine PH_3		**Pyramidal**	**Polar**
Hydrogencyanide HCN	**H - C ≡ N:**	**Linear**	**Polar**
Diberyllium Oxide Be_2O	**$:\mathrm{Be}:\ddot{\mathrm{O}}::\mathrm{Be}$**	**v-shaped (bent)**	**Polar**

Question 30

Phenolphthalein is a diprotic acid molecule and has two different equilibrium situations which are sensitive to concentrations of OH^- (aq). In pH range less than or equal to 8.3 the fully hydrogenated form is in such high concentrations that no evidence of the first ionisation forming H^+(aq) and a pink coloured ion, is visible.

If the phenolphthalein molecule is written as H_2PhTh then the equilibrium equation at say pH 7 could be written as

$$\underset{\text{colourless}}{H_2PhTh} + H_2O \rightleftarrows \underset{\text{pink}}{HPhTh^-(aq)} + H_3O^+(aq)$$

(a) From the information given above, would you expect the K value for the equation to be large or small?

The K value must be a lot less than 1.0.

[1 mark]

(b) If 5 ml of 0.1molL^{-1} NaOH is added to a neutral solution of phenolphthalein:

(i) What will be observed? **Pink colour appears. (1)**

(ii) Using Le Chatelier's Principle, explain why.

Adding hydroxide neutralises the H_3O^+, reducing it's concentration. (½)
The reaction proceeds to the right. (½)
To compensate for the change by increasing it's concentration. (½)
In doing so, it produces the pink ion/re-establishes equilibrium. (½)

[3 marks]

Question 31

(a) HCl is a strong acid. Calculate the pH of a solution of 0.001 mol L^{-1} HCl (aq)

[1 mark]

pH = - log [H^+] = $\underline{3.00}$

(b) Explain why the value for the pH of 0.001 mol L^{-1} HF (aq) will be different to 0.001 mol L^{-1} HCℓ (aq).

HF is a weak acid, which ionises less than the strong acid HCℓ.

pH should be greater than 3

[2 marks]

(c) A 5.0×10^{-4} mol L^{-1} solution of trichloroethanoic acid ($Cℓ_3CCOOH$) has a pH of 3.3. Calculate whether this acid is a strong or weak acid. Explain your reasoning.

$[H^+] = 10^{-3.3} = 5.01 \times 10^{-4}$ molL^{-1} (1)

$Cℓ_3CCOOH \rightarrow Cℓ_3CCOO^- + H^+$

(1)

5.01×10^{-4} molL^{-1} $\rightarrow$ 5.01×10^{-4} molL^{-1}

Fully ionised, therefore STRONG (1)

[3 marks]

(d) Write an equation for the ionisation of trichloroethanoic acid in water.

$Cℓ_3CCOOH(aq) + H_2O \rightarrow Cℓ_3CCOO^-(aq) + H_3O^+(aq)$

[1 mark]

(e) Estimate a pH value for a solution of $Cl_3CCOONa_{(aq)}$.

7.0

[1 mark]

(f) Calculate the pH of a mixture resulting from the addition of 250 ml of 0.10molL^{-1} $Ba(OH)_2$ and 750 ml of 0.050molL^{-1} HCl.

[3 marks]

Moles OH^- = 2 x (0.1 x 0.250) = 0.050mol

Moles H^+ = 0.05 x 0.75 = 0.0375mol

Moles OH^- left = 0.05 – 0.0375 = 0.0125mol

$[OH^-]$ = 0.0125/1L = 0.0125 molL^{-1}

$[H^+]$ = 10^{-14} / 0.0125 = 8×10^{-13} mol L^{-1}

pH = 12.1

Question 32

The melting points for five of six substances are shown in the table below arranged in their groups and taken from two periods in the Periodic table.

	GROUP	2	GROUP	14	GROUP	17
	$MgC\ell_2$	$BaC\ell_2$	$SiC\ell_4$	$PbC\ell_4$	$C\ell_2$	$IC\ell$
Melting Point (^oC)	710	860	20	500	—80	?

(a) Use suitable chemistry principles to explain why $PbC\ell_4$ has a similar melting point to $BaC\ell_2$ and $SiC\ell_4$ is closer to $C\ell_2$.

[2 marks]

$PbC\ell_2$ behaves as an ionic solid like the two in group 2. (1)

$SiC\ell_4$ behaves as a covalent molecular substance, like $C\ell_2$ (1)

(b) What is the major type of intermolecular bonding present in $SiC\ell_4$ and $IC\ell$ and which has the strongest bonding?

$IC\ell$ has strongest

In $SiC\ell_4$ only dispersion forces are responsible (1)

In $IC\ell$ dipole-dipole forces are the major force. (1)

[3 marks]

(c) Would you expect the melting point of $IC\ell$ to be **+ 50^{O}C** or **— 50^{O}C**? Explain why.

PLUS 50 it should be greater than the non-polar $SiCl_4$ and significantly different from the non-polar Cl_2.

[2 marks]

Question 33

Water ionises according to the equation

$$H_2O\ (l) \rightleftharpoons H^+\ (aq)\ +\ OH^-\ (aq)$$

(a) Write the equilibrium constant expression for this reaction here:

$$\mathbf{K = [H^+][OH^-]}$$ (1 mark)

At 25 °C, $[H^+] = [OH^-] = 1.0 \times 10^{-7}$ mol L^{-1}, and $K = 1.0 \times 10^{-14}$ $mol^2\ L^{-2}$.

At 50°C, the K value changes to approximately 5.5×10^{-14} $mol^2\ L^{-2}$.

(b) Use the information above, and Le Châtelier's principle, to predict whether the self-ionisation of water is an endothermic or exothermic process. Explain. (2 marks)

At 50 °C, the value for K_w increases, (½)
suggesting $[H^+] = [H_3O^+] = [OH^-]$ increases. (½)
This indicates that the reaction shifts to the right, (½)
suggesting that the reaction must be endothermic, as the system has moved to consume heat. (½)

(c) Calculate the pH of a sample of water at 50 °C. (2 marks)

$K_w = 5.5 \times 10^{-14} = [H_3O^+][OH^-]$

Neutral solution $\therefore$ $[H_3O^+] = [OH^-] = \sqrt{5.5 \times 10^{-14}} = 2.34 \times 10^{-7}$ mol L^{-1}

pH = - log (2.34×10^{-7})

= <u>6.63</u>

Question 34

An equilibrium is set up in a closed system held at 600°C, which started with some chips of calcium cyanamide, $CaCN_2$, and water. The equilibrium established is outlined by the **unbalanced** equation below:

$$CaCN_2(s) + H_2O(g) \rightleftharpoons NH_3(g) + CaCO_3(s)$$

Complete the following table, giving your answers as “increases’, “decreases” or no change”.

Change made to the Equilibrium system.	Effect on the rate of the forward reaction.	Effect on the equilibrium yield of $NH_3(g)$.
More water is added to the system.	**INCREASE**	**INCREASE**
A catalyst is added to the system	**INCREASE**	**NO CHANGE**

[4 marks]

Question 35

The bulk of the world’s nitric acid is made by passing ammonia and excess air through multilayered very fine rhodium-platinum gauze pad at a temperature of around 750°C. There is a sequence of purely gaseous reactions that must occur in the order that follows

$$4\,NH_3 + 5\,O_2 \rightleftharpoons 4\,NO + 6\,H_2O$$
$$2\,NO + O_2 \rightleftharpoons 2\,NO_2$$
$$3\,NO_2 + H_2O \rightleftharpoons 2\,HNO_3 + NO$$

(a) What is the purpose of the rhodium-platinum alloy present in the reacting chamber?

ACTS AS A CATALYST Speeds up the reaction rate [2 marks]

(b) Why is the gauze pad “multilayered and very fine”?

State of subdivision increases surface area and contact for the reactants [2 marks]

(c) If these reactions are carried out at atmospheric pressure a 50-55% yield results for the conversion of ammonia to nitric acid. Suggest whether increasing or decreasing the pressure of the closed system would be beneficial in improving the yield. Give reasons.

Overall reaction:

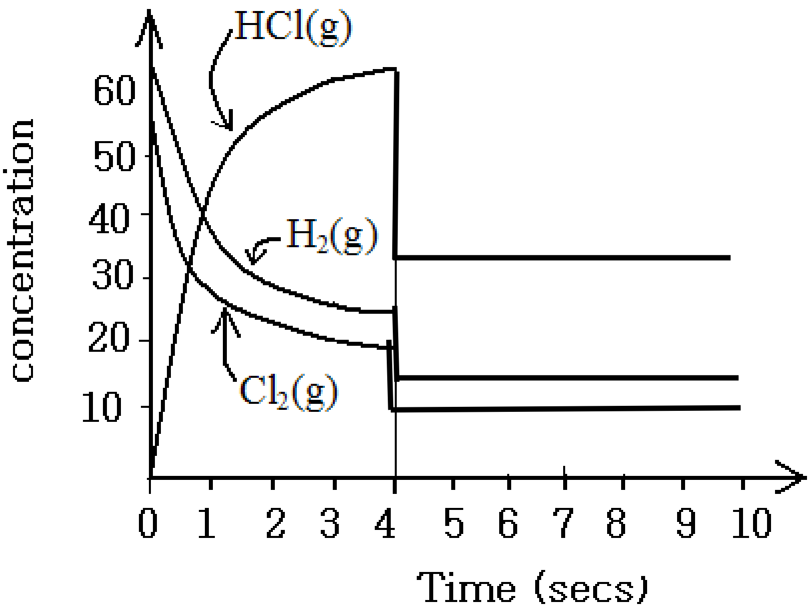
$$\mathbf{12\,NH_3 + 21\,O_2 \rightleftharpoons 8\,HNO_3 + 14\,H_2O + 4\,NO}$$

33 moles of gas LHS compared to 26 moles gas RHS, so pressure reduces in forward direction hence INCREASE PRESSURE OF SYSTEM, equilibrium shifts forward to oppose the change.

[2 marks]

Question 36

Consider $H_2(g) + Cl_2(g) \rightleftharpoons 2\,HCl(g)$ at $100^\circ C$



(a) For the system represented in the graph above, describe what is happening from the start to 4 seconds. (Assume that equilibrium is established after 4 seconds).

(2 marks)

H_2 an Cl_2 are reacting to produce HCl. (½)
As the reaction proceeds, rate of forward reaction reduces as [ ] of reactants decreases (½)
and the rate of reverse reaction increases as [ ] of products is increasing, (½) until equilibrium is achieved. (½)

(b) The volume of the system is doubled at 7 seconds. On the graph above, draw in the concentrations of all these gases from 4 seconds up to 10 seconds, by which time equilibrium is re-established. (3 marks)

Section Three: Extended answer 40% (80 Marks)

Question 37 **(16 marks)**

When bromine is added to cold aqueous sodium hydroxide solution, it loses colour due to the reaction

$$2NaOH(aq) + Br_2(l) \rightarrow NaBr(aq) + NaOBr(aq) + H_2O(l)$$

a) What mass of sodium hypobromite, NaOBr, can be produced by the reaction of 10.0g of bromine with 100.0 ml of 2.00 $molL^{-1}$ sodium hydroxide.

[8 marks]

b) Assuming the addition of the bromine has no significant effect on the volume of the final mixture, what is the pH of the final solution?

[5 marks]

c) What volume of 0.1500 $molL^{-1}$ silver nitrate solution is required to completely precipitate the sodium bromide produced in the reaction above?

[3 marks]

(a) n (Br_2) = 10.0 / (2 x 79.9) = 0.0626 moles [2 marks]

n (NaOH) = c x v = 2 x 0.1 = 0.200 moles [2 marks]

The Br_2 is the limiting reagent as it reacts with 2 x 0.0626 = 0.1252 moles of NaOH and leaves some in excess. (or similar reasoning) [1 mark]

n (NaOBr) = n (Br_2) = 0.0626 moles M[NaOBr] = 118.89 g

m (NaOBr) = n x M = <u>7.44 g</u> [3 marks]

(b) n (OH^-) = 0.200 - (0.0626 x 2) = 0.0748 moles [2 marks]

[OH^-] = n/v = 0.0748 / 0.100 = 0.748 [1 mark]

[H^+] = 10^{-14} / 0.748 = 1.337 x 10^{-14} [1 mark]

pH = - $\log_{10}[H^+]$ = <u>13.9</u> [1 mark]

(c) $Ag^+(aq)$ + $Br^-(aq)$ → AgBr(s) [1 mark]

0.1500 M 0.0626 moles (n(Br_2)) [1 mark]

n (Ag^+) = n (Br^-)

Vol (Ag^+) = 0.0626 / 0.15 = <u>0.417 litres or 417 ml</u>. [1 mark]

Question 38 **(10 marks)**

Ethanol (C_2H_5OH) can be used as an alternative to petrol. It burns in oxygen to form carbon dioxide and water.

Petrol can be considered pure octane for this question and the equation for its complete combustion is:

$$2C_8H_{18} + 25O_2 \longrightarrow 16CO_2 + 18H_2O$$

(a) Write an equation for the complete combustion of ethanol in oxygen.
[1 mark]

(b) Calculate the volume of carbon dioxide, at 25^{O}C and 200kPa, would be produced from the combustion of 60.0 g of ethanol.
[5 marks]

(c) Which would require the most oxygen for complete combustion: 60.0 g of ethanol or 60.0 g of petrol?
[4 marks]

(a) $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$ **[1 mark]**

(b) $n(C_2H_5OH) = 60.0 / 46.068 = 1.302$ mol **[2 marks]**
$n(CO_2) = (2/1) \times 1.302$ mol $= 2.6048$ mol **[1 mark]**
$V(CO_2) = nRT/P = 2.6048 \times 8.31 \times 298.15 / 200$ **[1 mark]**
$= \underline{32.3\ L}$ **[1 mark]**

(c) $n(C_2H_5OH) = 1.302$ mol
$n(O_2) = (3) \times n(C_2H_5OH) = \underline{3.906 \text{ mol of } O_2}$ **[1 mark]**

$n(C_8H_{18}) = 60.0 / 114.22 = 0.52528$ mol **[1 mark]**
$n(O_2) = (25/2) \times n(C_8H_{18}) = \underline{6.566 \text{ mol of } O_2}$ **[1 mark]**

Therefore the <u>petrol</u> requires more oxygen for its combustion [1 mark]

Question 39 **(16 marks)**

The green pigment verdigris, $Cu(CH_3COO)_2.H_2O$, is synthesed in three steps:

A 8.30g of $CuSO_4.5H_2O$ is dissolved in 100 ml water and heated. Aqueous ammonia is added dropwise until the deep blue colour of the $[Cu(NH_3)_4]^{2+}$ complex can clearly be seen. The mixture is diluted with water and the precipitate collected. The overall process can be shown as follows:

$$\mathbf{4CuSO_4.5H_2O_{(s)} + 6NH_4OH_{(aq)} \rightarrow CuSO_4.3Cu(OH)_{2(s)} + 3(NH_4)_2SO_{4(aq)} + 5H_2O_{(l)}}$$

B The precipitate is then added to water and stirred while 16.7 mL of 2.00 mol L^{-1} $NaOH_{(aq)}$ solution is added. The resulting precipitate of copper(**II**) hydroxide is filtered off and dried.

$$\mathbf{CuSO_4.3Cu(OH)_{2(s)} + 2NaOH_{(aq)} \rightarrow 4Cu(OH)_{2(s)} + Na_2SO_{4(aq)}}$$

C The precipitate from step **B** is mixed with excess pure ethanoic acid, and stirred until it is fully homogeneous.

$$\mathbf{Cu(OH)_{2(s)} + 2\ CH_3COOH_{(l)} \rightarrow Cu(CH_3COO)_2.H_2O_{(s)} + H_2O_{(l)}}$$

(a) Calculate the number of moles of $CuSO_4.3Cu(OH)_{2(s)}$ produced in step **A.** [3 marks]

(b) Determine the limiting reagent in step **B**. [5 marks]

(c) Calculate the mass of verdigris produced in step **C** [4 marks]

(d) Given that the density of pure ethanoic acid is 1.049 g mL^{-1}, what would be the minimum volume of ethanoic acid required in step **C** ? [4 marks]

(a) **$n(CuSO_4.5H_2O)$ = 8.30 / 249.69 = 0.03324 mol [2 marks]**
$n(CuSO_4.3Cu(OH)_2)$ = (1/4) x 0.03324 = <u>8.31×10^{-3} mol</u> [1 mark]

(b) **n(NaOH) = 2.00 x 0.0167 = 0.0334 mol [2 marks]**

$n(CuSO_4.3Cu(OH)_2)$ = 8.31×10^{-3} mol
if 8.31×10^{-3} mol of $CuSO_4.3Cu(OH)_2$, required n(NaOH) = (2/1) x 8.31×10^{-3}
= 0.0166 mol [1 mark]
Therefore NaOH is in excess (there are 0.034 mol),
so <u>$CuSO_4.3Cu(OH)_2$ is limiting reagent</u> [2 marks]

(c) **$n(Cu(OH)_2)$ produced in step B = (4/1) x $n(CuSO_4.3Cu(OH)_2)$**
= (4/1) x 8.31×10^{-3}
= 0.03324 mol [1 mark]

$n(Cu(CH_3COO)_2.H_2O)$ = (1/1) x 0.03324 = 0.03324 mol [1 mark]
$m(Cu(CH_3COO)_2.H_2O)$ = 0.03324 x 199.654 = <u>6.64 g</u> [2 marks]

(d) **$n(CH_3COOH)$ required = (2/1) x 0.03324 = 0.06648 mol [1 mark]**
$m(CH_3COOH)$ required = 0.06648 x 60.052 = 3.992 g [2 marks]
density = m(g) / volume (mL)
$\therefore$ volume = m / density = 3.992 /1.049 = <u>3.81 mL</u> [1 mark]

Question 40 **(14 marks)**

A sample of a mixture of pure sodium sulphate decahydrate ($Na_2SO_4.10H_2O$) and pure copper sulphate pentahydrate ($CuSO_4.5H_2O$) is dissolved in water and after thorough stirring, the solution is exactly halved. To one portion excess barium chloride is added, resulting in the formation of 10.740g of barium sulphate precipitate. Sufficient sodium hydroxide is added to the other portion to effect precipitation of insoluble hydroxides. The precipitate is filtered off and strongly heated to yield 1.591g of a black residue, CuO.

a. What is the total number of moles of sulfate ions present in the half portion which had the barium chloride added? [2 marks]

b. What is the number of moles of copper present in the original sample? [3 marks]

c. What is the mass of copper sulfate pentahydrate in the original sample? [3 marks]

d. How many moles of sodium sulfate decahydrate were present in the original sample? [3 marks]

e. What was the total mass of the original sample? [3 marks]

(a) All sulfate ions + $BaCl_2$ solution form $BaSO_4(s)$
M [$BaSO_4$] = 233.36
n ($BaSO_4$) = 10.74 / 233.36 = <u>0.0460 moles</u>. **[2 Marks]**

(b) M[CuO] = 79.55
n (CuO) = 1.591 / 79.55 = 0.0200 moles, in the half measure [2 Marks]
so <u>0.0400 moles</u> in the original sample **[1 mark]**

(c) n ($CuSO_4 . 5H_2O$) = n (CuO) and M[$CuSO_4 . 5H_2O$] = 249.69 [2 Marks]
m ($CuSO_4 . 5H_2O$) = 0.04 x 249.69 = <u>9.99 g</u> in the original sample. [1 mark])

(d) Total moles of sulfate in original sample = 0.0460 x 2 = 0.920 mol **[1 mark]**
The number of sulfate ions associated with the $Na_2SO_4 .10H_2O$ is the difference
0.092 – 0.040 = n($Na_2SO_4 .10H_2O$) = <u>0.052 moles</u> **[2 Marks]**

(e) M [$Na_2SO_4 .10H_2O$] = 322.2 g
mass of $Na_2SO_4 .10H_2O$ = 0.052 x 322.2 = 16.75 g **[2 Marks]**

Mass of sample taken = 16.75 + 9.99 = <u>26.74 g</u> **[1 mark]**

Question 41 (9 marks)

(a) A chemist needs to adjust the pH of a 25 000 L batch of waste water discharge that had a pH of 11.3 before releasing it into a stream. He estimated that he could do this by the addition of 420 L of 0.119 mol L^{-1} $HC\ell$ solution. Determine the pH of the batch of waste water that was discharged into the stream.
($K_w = 1 \times 10^{-14}$)

[7 marks]

$[H^+]$ = inverse log (–11.3)
= 5.012×10^{-12} $molL^{-1}$ **[1 mark]**

$[OH^-]$ = $\dfrac{1 \times 10^{-14}}{5.012 \times 10^{-12}}$

= 1.995×10^{-3} $molL^{-1}$ **[1 mark]**

moles of OH^- = 25 000 x 1.995×10^{-3}

= 4.988×10^{1} mol **[1 mark]**

moles of H^+ added = 420 x 0.119

= 4.998×10^{1} mol

moles of excess H^+ = 9.97×10^{-2} mol (0.1 mol) **[2 Marks]**

$[H^+]$ = $\dfrac{9.97 \times 10^{-2}}{(25000 + 420)}$

= 3.8×10^{-6} $molL^{-1}$ **[1 mark]**

pH = - log (3.8×10^{-6})

= <u>5.41</u> **[1 mark]**

(b) Consider the information in the table below. Is the pH of the treated waste water appropriate for release into a stream? Justify your answer.
(If you could not obtain a pH value in (a) above use a pH of 5.0)

Water source	pH
Acid rain	5.6
Ocean	8.8
Swan river at Armadale	7.9

[2 Marks]

- **The pH of the waste water is too low to be discharged into the water from an environmental viewpoint.**
- **The number of moles of excess acid is only small (0.099 moles) and given the volume of water and the resulting dilution in the river this would have minimal environmental impact.**
- **If the water was discharged into the alkaline river water it would be neutralised.**

Question 42 **(16 marks)**

Select a row (an example is Period 3) of the Periodic Table, and describe and explain the relationship between the number of valence electrons and an element's

i. bonding capacity
ii. position on the Periodic Table
iii. physical and chemical properties
iv. ionisation energy

Your answer should be approximately one to two pages in length.

Marks should be awarded for relevant chemistry, and the ability to write coherent sentences. It is not expected that all points listed below be covered, but each part of the question must be addressed for full marks to be awarded.

Diagrams, graphs and equations may be used to help where appropriate.
Some candidates may prefer to write 4 paragraphs, one for each part.
Other candidates may prefer to integrate the various parts of the question.

Points:
Bonding capacity

- **The number of valence electrons that an element can gain, lose or share in a bond**
- ***E.g.*** **Na has 1 valence electron, so it has a bonding capacity of 1; it tends to lose this to form a +1 ion.**
- **Mg has 2 valence electrons, so it has a bonding capacity of 2; it tends to lose these to form a +2 ion.**
- **S has 6 valence electrons, and it has a bonding capacity of 2 as it tends to gain two electrons to form a –2 ion.**
- **Cℓ has 7 valence electrons, and it has a bonding capacity of 1 as it tends to gain one electron to form a –1 ion.**
- **This results in compounds such as $NaC\ell$, $MgC\ell_2$, MgO**
- **Some students may discuss the high stabilities of the noble gases such as Ar.**

Position in the Periodic Table

- **Elements are placed in order of atomic number, which means in order of increasing number of electrons. Successive elements along a row have successively increasing numbers of valence electrons.**
- ***E.g.*** **Row 3 Na electron configuration 2,8,1**
  Mg electron configuration 2,8,2, etc
- **The number of valence (outer shell) electrons has a strong effect on the element's bonding, and therefore chemical, properties.**
- **Students may also comment on the elements on the left and the elements on the right and perhaps relate first ionisation energy to the metallic characteristics of the element.**

Physical and chemical properties

- **The number of valence electrons influences the element's physical and chemical properties**
- **Group 1 elements (*e.g.* Na 2,8,1) have one valence electron. Since this is weakly held, it can move under the influence of an electric field, and so conduct electricity. This loosely held valence electron is also responsible for the ability of metals like Na to conduct heat.**
- **Na tends to lose its one valence electron and form ionic bonds with non metals such as oxygen (which tends to gain 2 electrons to form a –2 ion)**

(Points similar to the above may also be discussed in relation to the non-metallic elements on the right of the Periodic Table.)

Ionisation Energy

- **Graph of first ionisation energy (IE) of elements from Na to Ar**
- **Na has one valence electron which is easily lost so Na has a relatively low first IE and a very high second IE**
- **Mg has two valence electrons; so Mg has a relatively low first IE, a higher second IE but a very high third IE**
- **Thus, first IE generally increases across a row of the Periodic Table**
- **Therefore the more valence electrons possessed by an element, the greater the first IE.**
- **Group 16 and 17 elements such as S and Cℓ are involved in covalent bonding because of their high first IEs.**

Mark	Description
12-16	Coherently describes and explains all parts, at least 15 points made, some with supporting examples and/or diagrams.
9-11	All parts clearly addressed but only 10-14 points made; or at least 15 points made but only three parts addressed; few examples or diagrams
6-10	5-9 points made but only three parts addressed; or 8-12 points made but only two parts addressed; no, or one or two, examples or diagrams
1-5	Maximum of four points in two parts addressed
0	Question incorrectly answered or not attempted

END OF EXAMINATION